

# PAGING TECHNIQUE

## C PROGRAM:

```
#include <stdio.h>

int main() {

    int pageTable[20];

    int pageSize;

    int numPages;

    int logicalAddress;

    int pageNumber;

    int offset;

    int frameNumber;

    int physicalAddress;

    int i;

    printf("Enter Page Size: ");

    scanf("%d", &pageSize);

    printf("Enter Number of Pages: ");

    scanf("%d", &numPages);

    printf("Enter Frame Numbers for Each Page:\n");

    for(i = 0; i < numPages; i++)

    {

        printf("Page %d -> Frame: ", i);

        scanf("%d", &pageTable[i]);

    }

    printf("Enter Logical Address: ");

    scanf("%d", &logicalAddress);

    pageNumber = logicalAddress / pageSize;

    offset = logicalAddress % pageSize;
```

```

if(pageNumber >= numPages)
{
printf("Invalid Logical Address\n");
return 0;
}

frameNumber = pageTable[pageNumber];

physicalAddress = (frameNumber * pageSize) + offset;

printf("\nPage Number : %d", pageNumber);

printf("\nOffset : %d", offset);

printf("\nFrame Number : %d", frameNumber);

printf("\nPhysical Address : %d\n", physicalAddress);

return 0;
}

```

## SHELL SCRIPTING:

```

#!/bin/bash

echo "Enter Page Size:"

read pageSize

echo "Enter Number of Pages:"

read numPages

declare -a pageTable

for ((i=0;i<numPages;i++))

do

echo "Enter Frame Number for Page $i:"

read pageTable[$i]

done

echo "Enter Logical Address:"

read logicalAddress

pageNumber=$((logicalAddress / pageSize))

```

```
offset=$((logicalAddress % pageSize))  
if [ $pageNumber -ge $numPages ]  
then  
echo "Invalid Logical Address"  
exit  
fi  
frameNumber=${pageTable[$pageNumber]}  
physicalAddress=$((frameNumber * pageSize + offset))  
echo "Page Number : $pageNumber"  
echo "Offset : $offset"  
echo "Frame Number : $frameNumber"  
echo "Physical Address : $physicalAddress"
```

OUTPUT:

```
(kali㉿ Anbuchelvan)-[~]  
$ nano paging.sh  
  
(kali㉿ Anbuchelvan)-[~]  
$ chmod +x paging.sh  
  
(kali㉿ Anbuchelvan)-[~]  
$ bash paging.sh  
Enter Page Size:  
2  
Enter Number of Pages:  
3  
Enter Frame Number for Page 0:  
1  
Enter Frame Number for Page 1:  
2  
Enter Frame Number for Page 2:  
3  
Enter Logical Address:  
4  
Page Number : 2  
Offset : 0  
Frame Number : 3  
Physical Address : 6
```